8 Fig. 8.1 shows how the electromotive force (e.m.f.) of a 60 Hz alternating current (a.c.) power supply varies with time.

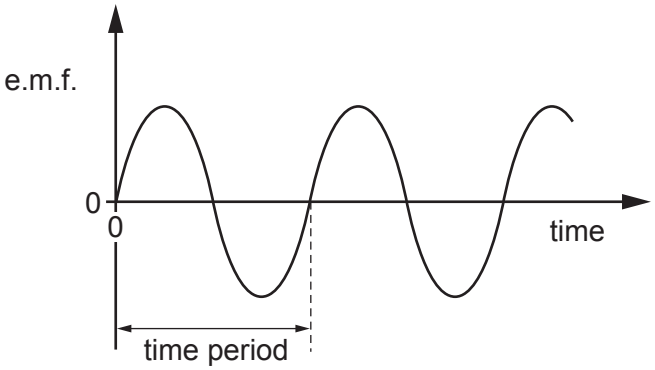


Fig. 8.1

(a) Calculate the time period of the a.c.

time period = ... [1]

(b) Fig. 8.2 shows this power supply connected in a circuit.

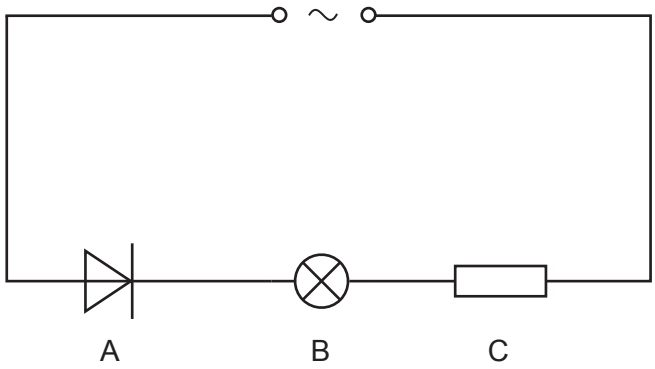


Fig. 8.2

(i) State the name of component A.

..... [1]

(ii) In each time period of the a.c., 1.5×10^{17} electrons pass through component A. The charge on an electron is 1.6×10^{-19} C.

Calculate the average current in the circuit during one time period.

current = ... [3]

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(c) On Fig. 8.3:

1. mark, with an arrow labelled E, the direction of the electron flow through component B

2. mark, with an arrow labelled I, the direction of the conventional current in component C.

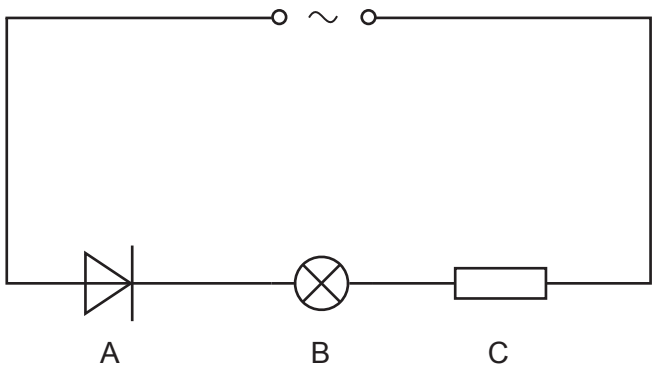


Fig. 8.3

[2]

(d) Fig. 8.4 shows a circuit with components B and C connected to a direct current (d.c.) power supply of e.m.f. 12 V.

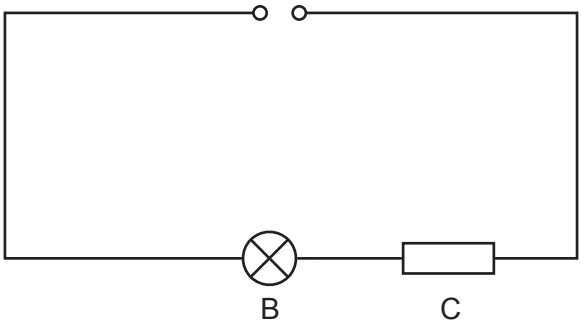


Fig. 8.4

The current in the circuit is 0.35 A.

Calculate the power delivered by the power supply to the circuit.

power = ... [2]

[Total: 9]